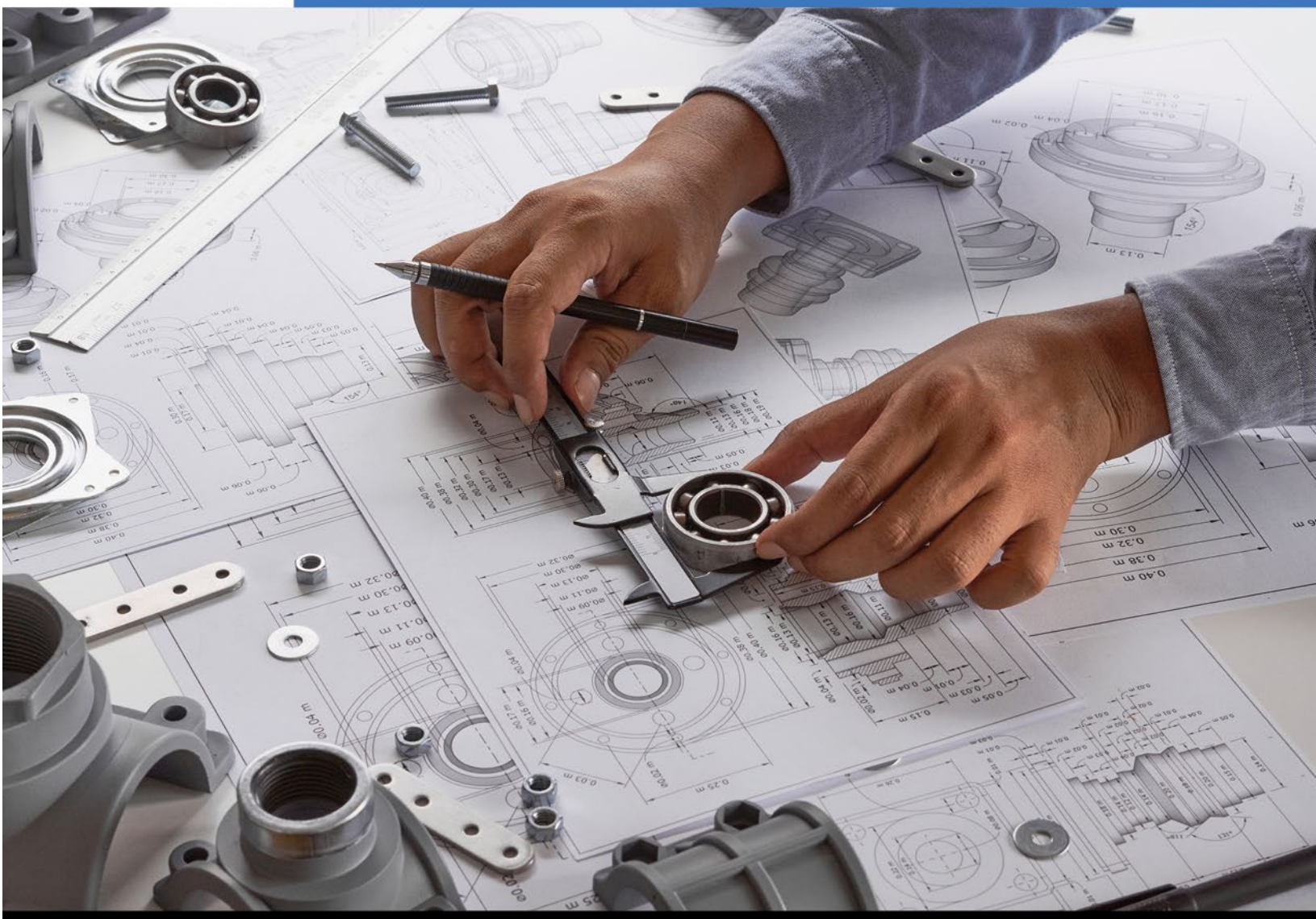
SD-10

Guide for Identification and Development of Metric Standards



DEFENSE STANDARDIZATION PROGRAM OFFICE

January 2026

STDZ

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Foreword

This guide helps organizations in the Department of War (DoW) decide when it is necessary to create metric specifications and standards, and the best way to express metric units of measurement in those documents. The availability of metric documents is a prerequisite to the creation of metric-based weapon systems and equipment. This document helps DoW program offices, standardization management activities, engineering offices, contractors, and others meet the requirements of Section 5164, *Metric Usage*, of Public Law 100-418, the Omnibus Trade and Competitiveness Act of 1988.

The guide updates the information in the May 2019 version of SD-10. This document is approved for public release. Electronic copies are available from ASSIST at https://assist.dla.mil or from the Defense Standardization Program Office website at https://www.dsp.dla.mil.

If you have any questions or comments about this document or metric use in DoW, please contact the Defense Standardization Program Office, 8725 John J. Kingman Road, Stop 5100, Fort Belvoir, VA 22060-6220 or email DSPO-Director@dla.mil.



MICHAEL A. HEAPHY JR.
Director, Defense Standardization Program Office

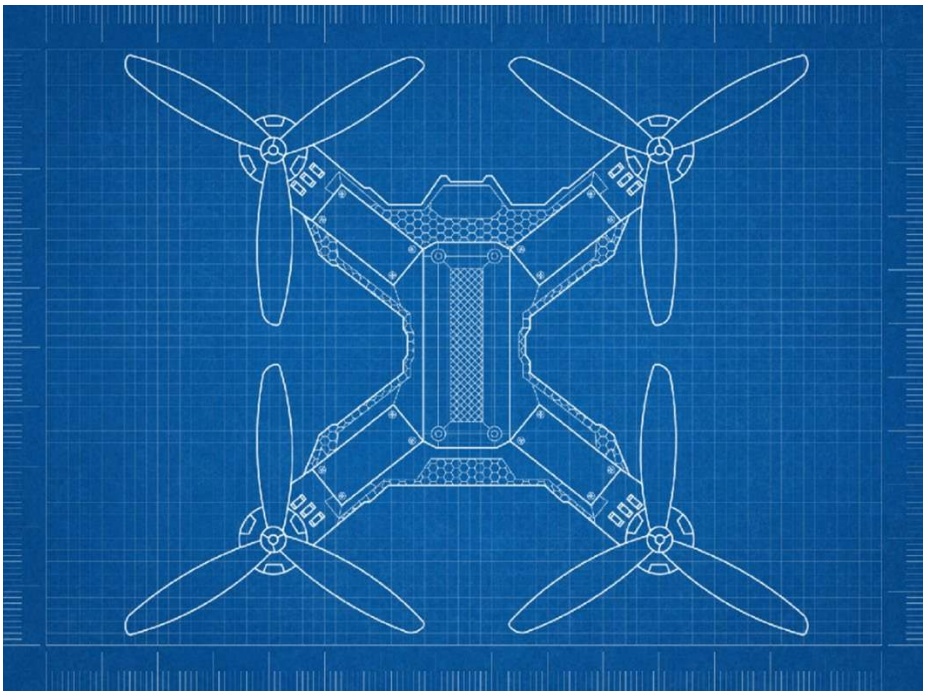
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Section 1. Scope

This document supplies guidance for the identification and development of standards and specifications using the International System of Units (SI), commonly known as the metric system.[1] The guide does not cover metric practice, such as methods of converting and rounding, which is addressed in Institute of Electrical and Electronics Engineers (IEEE)/ASTM SI 10, *American National Standard for Metric Practice*.

Section 2. Background

The Metric Conversion Act of 1975 (15 U.S.C. § 205(b)(1)) states that it is the "policy of the United States to designate the [SI] as the preferred system of weights and measures for United States trade and commerce."[2] In addition, 15 U.S.C. § 205(b)(2) requires each federal agency to "use the metric system of measurement in its procurements, grants, and other business-related activities, except to the extent that such use is impractical or is likely to cause significant inefficiencies or loss of markets to United States firms, such as when foreign competitors are producing competing products in" U.S. customary units.[3] Executive Order 12770, issued in 1991, further implemented the Metric Conversion Act, designating the Secretary of Commerce as the director and coordinator of conversion efforts by federal departments and agencies.[4]



[1] Since 1960, it has been known as the International System of Units (*The International System of Units*, 9th ed., Bureau International des Poids et Mesures, 2019.)
[2] Metric Conversion Act of 1975, 15 U.S.C. § 205(b)(1). https://www.govinfo.gov/link/uscode/15/205b.
[3] Metric Conversion Act of 1975, 15 U.S.C. § 205(b)(2). https://www.govinfo.gov/link/uscode/15/205b.
[4] Executive Order 12770, *Metric Usage in Federal Governmental Programs*. https://www.federalregister.gov/executive-order/12770.

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Section 3. Definitions

3.1 Conversion

Conversion is the process of changing a measurement from U.S. customary units to equivalent metric units within acceptable measurement tolerances. The physical configuration of the item is not changed.

3.2 Hybrid Specification

A hybrid specification is one in which some requirements are given in rounded, rational metric units, and other requirements are given in rounded, rational U.S. customary units. Hybrid specifications are often required for new designs where existing usable components must interface in a metric system.

3.3 Metric Specification

Metric specifications have requirements given in rounded, rational metric units, usually from being developed in metric units. The magnitudes are meaningful and practical (for example, 10 grams, not 0.35273 ounces). Documents containing only electrical units in both the metric and U.S. customary systems (e.g., volts, amps, and ohms) are classified as metric documents. Documents containing dimensional interfaces must have these interfaces in metric sizes to be classified as metric documents. Metric specifications are created for items to interface or operate with other metric items.

3.4 Metric System (SI)

15 U.S.C. § 205(c)(4) defines the metric system of measurement as "the International System of Units established by the General Conference of Weights and Measures"[5] ... "as interpreted or modified for the United States by the Secretary of Commerce."[6] The Secretary of Commerce delegated the responsibility for interpretation or modification to the National Institute of Standards and Technology (NIST). The official interpretation of the SI for the United States is published as a *Federal Register* notice[7] and in two companion guides, NIST Special Publication (SP) 330, *The International System of Units (SI)*, and SP 811, *Guide for the Use of the International System of Units (SI)*.[8] NIST continues to periodically interpret metric units and publish updates in the *Federal Register*.

3.5 Metric units

3.5.1 Metric units are a system of basic measures defined by the International System of Units in the SI brochure, "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and

[5] For information on the General Conference of Weights and Measures, see https://www.bipm.org/en/committees/cg/cgpm.
[6] Metric Conversion Act of 1975, 15 U.S.C. § 205(c)(4). https://www.govinfo.gov/link/uscode/15/205c.
[7] "Notices: Interpretation of the International System of Units (the Metric System of Measurement) for the United States," *Federal Register*, Vol. 73, No. 96, May 16, 2008, issued by the National Institute of Standards and Technology, Department of Commerce: https://www.federalregister.gov/documents/2008/05/16/E8-11058/interpretation-of-the-international-system-of-units-the-metric-system-of-measurement-for-the-united.
[8] For additional information on NIST SP 330 and SP 811 and the SI, contact: National Institute of Standards and Technology, Office of Weights and Measures-Laws and Metric Program, 100 Bureau Drive Stop 2600, Gaithersburg, MD 20899-2600, by email at TheSI@nist.gov, or by phone at (301) 975-3690. You will find the Laws and Metric Program's website at http://www.nist.gov/metric.

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Measures (BIPM).[9] These are also commonly called metric system units and described in IEEE/ASTM SI 10.[10]

3.5.2 In the United States, the authoritative SI source document is NIST SP 330. For use in the United States, this is the official English language translation approved by the BIPM of its publication, "Le Systeme International d'Unites." Guidance for use of NIST SP 330 is furnished in NIST SP 811.

3.6 Not Measurement Sensitive Document

A not measurement sensitive document is one in which application of the requirements does not depend substantially on some measured quantity. This type of document can use the metric system or the U.S. customary measurement system.

3.7 Substitution

Substitution is the process of changing a measurement from U.S. customary units to non-equivalent metric units, which necessitates changes to the item outside those permitted by established measurement tolerances. Requirements are given in rounded, rational metric units. A new metric item requiring a new part identification is created to eventually replace the U.S. customary version of the item.

3.8 U.S. Customary Specification

U.S. customary specifications have requirements given in rounded, rational U.S. customary units, usually from being developed in U.S. customary units. The magnitudes are meaningful and practical (for example, 1 ounce, not 28.3495 grams). U.S. customary specifications should include those with rounded, rational U.S. customary units only. Sometimes, specifications may use metric units that are the result of mathematical conversion from rounded, rational U.S. customary units. Such specifications should be considered U.S. customary documents since the metric conversions do not reflect requirements given in rounded, rational metric units. U.S. customary specifications are created for items to interface or operate with other items that are specified in U.S. customary units.

3.9 U.S. Customary Units

U.S. customary units are a system of measurements historically used in the United States, previously called inch-pound units.

Section 4. When and How Should Metric Standards and Specifications Be Used?

Metric practice is most economical and practical when new products and processes are designed. When deciding to create a new metric document or to convert an existing document from U.S. customary to metric, it is important to understand the purpose, assess the value, and consider the overall effects as well as the following:

- Common measurement practice of and effect on the relevant industrial base

[9] For more information about BIPM, see https://www.bipm.org.
[10] IEEE/ASTM SI 10, *American National Standard for Metric Practice*, is available from https://standards.ieee.org or https://astm.org.

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- Effect on the supplier base and supply availability
- Need for compatibility with foreign equipment
- Need for international standardization
- Item or process criticality and associated risk of failure
- Requirements for interface with other equipment and components
- Life expectancy of the item or process
- Overall cost, including modifications to transportation equipment, support equipment, documentation, training, and disposition of existing inventory.



4.1 When Should Conversion Be Used?

If the purpose is simply to put the requirements in metric language, conversion is appropriate. The conversion can state only the converted metric units in the requirements in lieu of the U.S. customary units, state the metric units in parentheses after the U.S. customary units in the requirements or vice versa, or give a table of conversions or conversion factors with the requirements in only one system. Conversion should be used in the following circumstances:

a. The technologies addressed in the document are based on the U.S. customary system internationally. Examples include most areas of electronic packaging and grid spacing, hydraulics, and tire rim sizes. Since the purpose of metric conversion is usually to achieve international standardization or interoperability, rather than metrication per se, there is no need for substitution in these areas. If inclusion of metric units is desired, conversion is appropriate.

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b. The items specified have no need for interoperability, interfacing, or international standardization. Examples include standalone items, such as articles of clothing and furniture.

c. The items or processes are not usually specified using rounded, rational numbers in either system. In these cases, economic and technical considerations are balanced to achieve an optimal situation. Examples include anti-corrosion coating thicknesses, electrical insulation thicknesses, locations of components in electronic assemblies, etc.

d. The items are expressed as capacity or volume and the system of measurement has no effect on the end use. Examples include fuels, paints, and other coatings. Metric hardware or equipment can be painted using gallons or liters of paint and can be powered by gallons or liters of fuel.

e. The documents are test methods or process standards. Adding equivalent metric dimensions permits the use of scales, micrometers, gauges, and other instruments calibrated in either system of units. However, if the existing standards and test methods affect final physical configuration or performance requirements, metric standards will require careful preparation.



4.2 When Is Substitution Needed?

Substitution is often needed to achieve international harmonization or to operate with metric hardware.

New metric design standardization documents are created for the following:

a. Items that have universal application and which require standardization to supply metric components for designers of metric equipment.

b. Items, such as wire sizes, screw thread forms, fasteners, tubing sizes, and other parts, used in new systems specified in metric dimensions.

c. Existing system designs that are changing to permit metric system standards; production machinery, raw materials, spare parts, and maintenance tools; and for which the most cost-effective set of such changes must be selected.

d. Items for which a reduction in the ranges, types, styles, classes, or products is facilitated by adoption of a family of metric items.

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e. Items that are peculiar to DoW and represent technological advances for operational application in new equipment.

f. Items that have a universal application of function and for which industry is designing the items in the metric system of measurement.

New metric standardization documents often supplement specifications and standards, not necessarily supersede them. As the use of the metric system grows, use of metric standardization documents will increase, while use of U.S. customary documentary standards will decrease.

4.3 When Is Conversion or Substitution Unnecessary?

Conversion and substitution are not necessary for all items, such as the following:

a. Items that are becoming obsolete, making the change to metric impractical.

b. Items which will continue in use without modifications. No useful purpose is served by translating inches to millimeters for such items, their spare parts, or the tools required to maintain them since production and maintenance will continue to use U.S. customary system facilities and standards.

c. Items of very limited applications where costs of metric modules outweigh the benefits.

d. Not measurement sensitive documents (see 3.6).

4.4 Metric Approaches

The following approaches should be considered when creating specifications and standards to support metric usage:

a. *New parallel document*. For very complex documents filled with many conversion-susceptible measurements, issue a new SI standardization document. Carefully prepare the new document with the substitution method and select sensible equivalents. Once the new SI standardization document is prepared, revise the basic document and the metric document concurrently until the U.S. customary document is no longer required and is cancelled.

b. *Metric appendix*. For less complex documents, or for very complex documents where retention of the original document number is considered necessary, a substitution appendix can be prepared. The basic document would remain in U.S. customary units and refer to the appendix for metric information. The appendix would refer to the basic document for technical features and cite only the carefully selected metric equivalents.

c. *Metric notes*. For relatively simple documents with only a few measurement units, metrication can be handled by appropriate notes or by one or more footnotes.

d. *Contract wording*. Metric requirements can be inserted in contracts in lieu of creating metric specifications or standards.

4.5 Metric Values

Two basic methods can be used to arrive at practical and meaningful metric values in conversion by substitution for individual quantity requirements in a document:

a. *Size substitution*. Simple replacement of standard U.S. customary size with the existing accepted metric size. This is often used to conform to internationally recognized metric modules. Use of preferred types and sizes and parts and materials, especially those of the International Organization for Standardization (ISO) and International Electrotechnical Commission (IEC), should be encouraged. Alteration of international or foreign standards to fit domestic needs before adoption is sometimes necessary.

b. *Adaptive conversion*. A change from a rounded, rational U.S. customary quantity to a rounded, rational metric quantity which is reasonably equivalent (as, for example, speed limits). The

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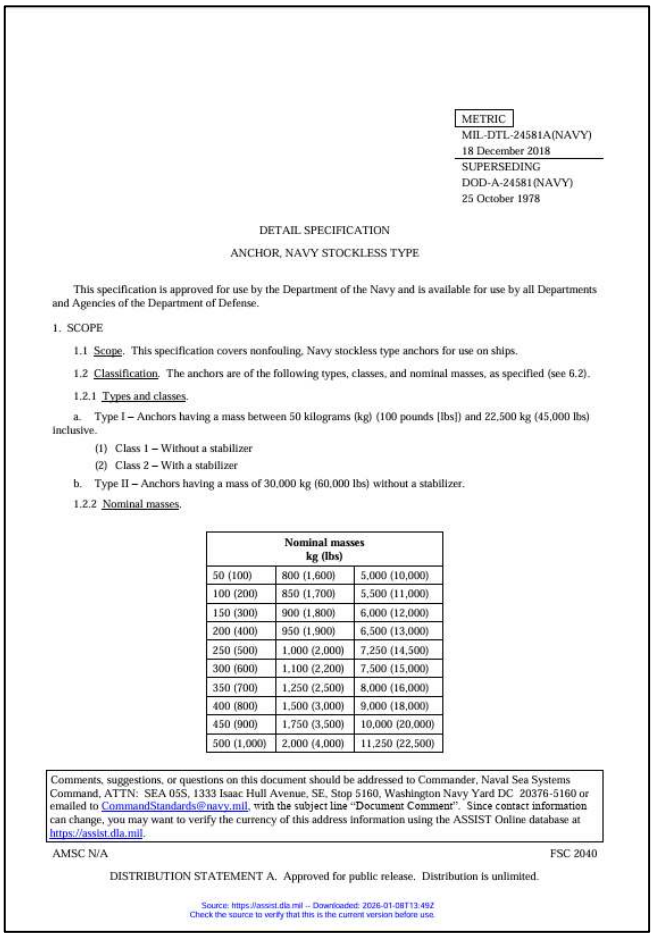
decision on which to use must be based on knowledge, experience, and common sense. It is important to address rationalization and variety reduction and to consider arithmetic or geometric series of sizes. It is desirable to use metric dimensions for physical interface in hybrid systems, except between U.S. customary items.

Section 5. Dual Dimensions

Avoid the use of both metric and U.S. customary measurements on drawings and other pictorial illustrations in a specification or standard. The use of tables to translate the specific U.S. customary units to metric units is acceptable. For text material, when preference is given to U.S. customary units, acceptable metric units may be shown in parentheses.

When preference is given to metric units, U.S. customary units may be omitted or included in parentheses. In general, where it has long been standard practice to cite metric units alone (such as citing temperatures only in degree Celsius), U.S. customary equivalents may be omitted. Only specify a repetitive equivalent (for example, 1 inch [25.4 mm]) the first time it appears in a paragraph of a standardization document.





METRIC
MIL-DTL-24581A(NAVY)
18 December 2018
SUPERSEDING
DOD-A-24581(NAVY)
25 October 1978

DETAIL SPECIFICATION

ANCHOR, NAVY STOCKLESS TYPE

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers nonfouling, Navy stockless type anchors for use on ships.

1.2 Classification. The anchors are of the following types, classes, and nominal masses, as specified (see 6.2).

1.2.1 Types and classes.

a. Type I – Anchors having a mass between 50 kilograms (kg) (100 pounds [lbs]) and 22,500 kg (45,000 lbs) inclusive.

(1) Class 1 – Without a stabilizer

(2) Class 2 – With a stabilizer

b. Type II – Anchors having a mass of 30,000 kg (60,000 lbs) without a stabilizer.

1.2.2 Nominal masses.

Nominal masses kg (lbs)		
50 (100)	800 (1,600)	5,000 (10,000)
100 (200)	850 (1,700)	5,500 (11,000)
150 (300)	900 (1,800)	6,000 (12,000)
200 (400)	950 (1,900)	6,500 (13,000)
250 (500)	1,000 (2,000)	7,250 (14,500)
300 (600)	1,100 (2,200)	7,500 (15,000)
350 (700)	1,250 (2,500)	8,000 (16,000)
400 (800)	1,500 (3,000)	9,000 (18,000)
450 (900)	1,750 (3,500)	10,000 (20,000)
500 (1,000)	2,000 (4,000)	11,250 (22,500)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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Section 6. Other Guidance Publications

The following publications supply additional guidance on metric use and practices.

Aerospace Industries Association-NAS 10000, *NA Documents Preparation and Maintenance in SI (Metric) Units*. Available from https://aia-aerospace.org.

Aerospace Industries Association-NAS 10001, *Preferred Metric Units for Aerospace*. Available from https://aia-aerospace.org.

American Petroleum Institute, *Guidelines for Use of the International System of Units (SI) in the Petroleum and Allied Industries*, "API Manual of Petroleum Measurement Standards," Chapter 15. Available from https://www.api.org/standards.

American Society of Agricultural and Biological Engineers EP285.8, *Use of SI (Metric) Units*. Available from https://www.asabe.org.

American Society of Mechanical Engineers (ASME), *ASME Orientation and Guide for Use of SI-1 (Metric) Units*. Available from https://www.asme.org.

ASME B4.2, *Preferred Metric Limits and Fits*. Available from https://www.asme.org.

ASME B4.3, *General Tolerances for Metric Dimensioned Products*. Available from https://www.asme.org.

American Welding Society A1.1, *Metric Practice Guide for the Welding Industry*. Available from https://www.aws.org.

Department of Commerce/NIST SP 330, *The International System of Units*. Available from https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.330-2019.pdf.

Department of Commerce/NIST SP 811, *The NIST Guide for the Use of the International System of Units*. Available from https://physics.nist.gov/cuu/pdf/sp811.pdf.

Department of Commerce/NIST SP 814, *Interpretation of the SI for the United States and Metric Conversion Policy for Federal Agencies*. Available from https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication814.pdf.

Department of Commerce/NIST, *Putting the Best "Foot" Forward: Ending the Era of the U.S. Survey Foot (1959 to 2022)*. Available from https://www.nist.gov/pml/us-surveyfoot.

General Services Administration PBS-PQ260, *Metric Design Guide*. Available at https://www.gsa.gov.

IEEE 260.1, *IEEE Standard Letter Symbols for Units of Measurement (SI Customary Inch-Pound Units, and Certain Other Units)*. Available from https://standards.ieee.org.

IEEE 945, *IEEE Recommended Practice for Preferred Metric Units for Use in Electrical and Electronics Science and Technology*. Available from https://standards.ieee.org.

IEEE/ASTM SI 10, *American National Standard for Metric Practice*. Available from https://standards.ieee.org or https://astm.org.

Interagency Committee on Metric Policy FED-STD-376, *Preferred Metric Units for General Use by the Federal Government*. Available from https://assist.dla.mil.

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National Aeronautics and Space Administration (NASA) Policy Directive 7120.4E Subject: NASA Engineering and Program/Project Management Policy. Available at https://nodis3.gsfc.nasa.gov.

Society of Automotive Engineers (SAE) AIR 1657, *Handbook of Hydraulic Metric Calculations*. Available from https://www.sae.org

SAE AIR 1758, *Limits and Fits—International Metric Tolerance System*. Available from https://www.sae.org.

SAE J390, *Dual Dimensioning*. Available from https://www.sae.org.

SAE TSB003, *Rules for SAE Use of SI (Metric) Units*. Available from https://www.sae.org.

Underwriters Laboratories (UL), *Metric Policy*. Available at https://www.ul.com.

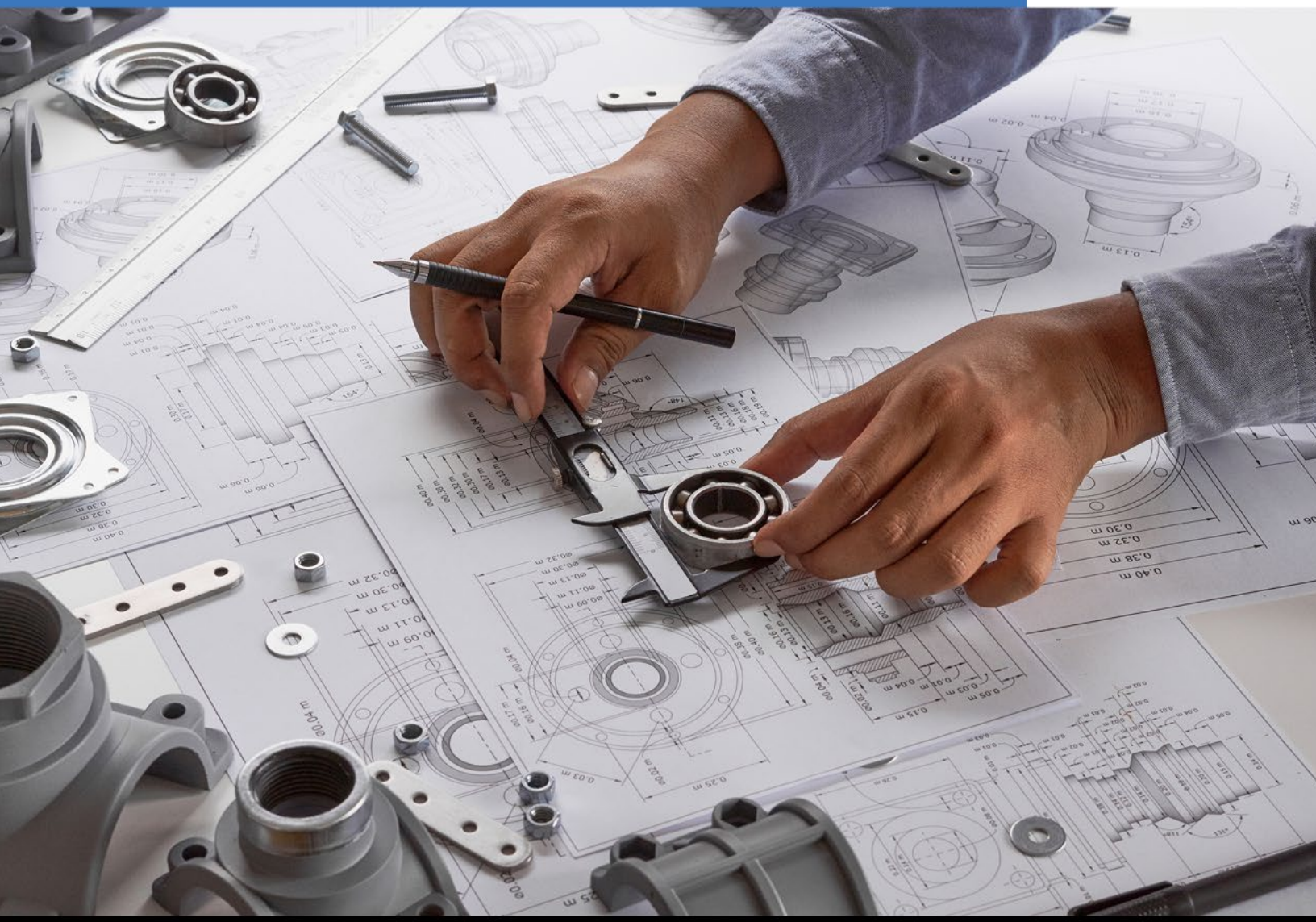
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Appendix A. Abbreviations

ASME	American Society of Mechanical Engineers
ASSIST	Acquisition Streamlining and Standardization Information System
BIPM	International Bureau of Weights and Measures
DoW	Department of War
IEEE	Institute of Electrical and Electronics Engineers
NIST	National Institute of Standards and Technology
SAE	Society of Automotive Engineers
SD	Standardization-related Document
SI	International System of Units
SP	Special Publication

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